

Technology and Social Change

William Fielding Ogburn Revisited

STEVEN L. DEL SESTO

ABSTRACT

This paper reviews William Fielding Ogburn's work on technology and social change. An overview of his basic theory is presented, as well as later refinements found in such works as *Recent Social Trends in the United States*, *Technological Trends and National Policy*, *The Social Effects of Aviation*, and other books and articles. It is suggested that while Ogburn is most widely remembered for his thesis of cultural lag, some of his later work on technology and social change is actually more important and deserves reconsideration and extension. One kind of extension might follow the lines of a systems approach rather than continuing the technological primacy strategy that Ogburn is normally considered to have used. The paper also surveys Ogburn's important work in the areas of technology, planning, and social policy, including materials on technology assessment. Ogburn's nontechnocratic approach to these questions, as well as his insistence that social scientists have a great deal to offer to the formulation of technological and social policies, are some of his most important contributions. Questions of technology and social change remain important contemporary issues, yet there is a surprising paucity of social science work on these crucial topics. Further study, extension, and reformulation of Ogburn's pioneering ideas offer fruitful means of embarking on new social science analyses of technology and society.

Introduction

In his classes at the University of Chicago, William Fielding Ogburn was fond of using the radio to illustrate the relationship between technology and social change. In 1922, when Ogburn was in the midst of writing his first important book, *Social Change, with Respect to Culture and Original Nature* [19], radios were scarce. As late as 1919 radio broadcasting did not exist. Some wireless sets were in use, and the "radiophone" was far enough advanced so that people flying over Manhattan office buildings could communicate with the occupants. Essentially, however, no music or speeches were broadcast, except for a few weak efforts. Not until the spring of 1920 did Frank Conrad of the Westinghouse Electric Corporation in Pittsburgh set up the world's first radio broadcasting station. By the spring of 1922 the radio had become a national craze, and the sale of radio sets, parts, and accessories amounted to more than \$60 million annually. By the end of 1929, sales had climbed to a remarkable \$843 million (a 14-fold increase over 1922 figures!), while Radio Corporation of America shares soared to record highs on the New York Stock Exchange [3].

STEVEN L. DEL SESTO is currently Visiting Scholar at the Russell Sage Foundation in New York. He is author of *Science, Politics, and Controversy: Civilian Nuclear Power in the United States, 1946-1974* and has written extensively on the social aspects of technology.

Address reprint requests to Professor Steven L. Del Sesto, Russell Sage Foundation, 112 East 64th Street, New York, New York 10021.

The radio was more than a passing national fad and an instrument for entertainment and diversion, however; it was also a great advance in communications that helped link all parts of the country into a mass society sharing common interests, values, ideas, and beliefs. Radio would become the molder of popular opinion and the creator of political support and consensus for national policies. In short, it was reshaping the social fabric. Ogburn saw that inventions like the radio offered important opportunities to study the wide-ranging social effects of technology.

Today, if Ogburn is mentioned at all, he is usually identified with the thesis of cultural lag, which is often criticized as an oversimplified, basically unoriginal version of the technological primacy argument of social change. Ogburn's other important work, meanwhile, is swept aside, largely ignored, or just forgotten despite the fact that his writings remain among the most important contemporary statements of the relationship between technology and social change.

The purpose of this paper is to revisit Ogburn's original theory, including later extensions found primarily in his contributions to the federal government's first national assessments, *Recent Social Trends in the United States* [20], issued in 1933, and *Technological Trends and National Policy, Including the Social Impact of New Inventions* [21], issued in 1937. Ogburn was the research director of *Recent Social Trends* as well as an influential member of the National Resources Committee, which produced *Technological Trends*. An appreciation of this work provides a good view of the "state-of-society" assumptions underlying the first national assessments and demonstrates the importance of social science in these efforts. Indeed, the kinds of assumptions made by Ogburn and his colleagues remain implicit in many current-day assessments. The final section of the paper points up some of the strengths and weaknesses of Ogburn's theories and methods and the lines along which further research might be pursued.

Technology and Social Change: Overview of the Early Theory

If Ogburn were alive today, he might well be expounding on the anticipated social effects of civilian nuclear power reactors, advances in computers and information technology, or the United States space program. In the 1920s and 1930s, however, Ogburn and his colleagues had their hands full with the radio, automobile, airplane, and other advances that were literally reshaping American life. The sheer magnitude of change prompted Ogburn and his associates to embark on a general analysis of the effects of technology on the American social structure.

In 1922 Ogburn took up this task in perhaps his most important book, *Social Change, with Respect to Culture and Original Nature* [19]. Much of this book refuted biological interpretations of society and social evolution and attempted to establish a more rigorous sociocultural explanation. Probably encouraged in part by his contacts in the Anthropology Department at Columbia University, where he spent his early career, Ogburn had intensively studied the growing body of ethnographic field work that convincingly showed that all human beings, even those living in primitive societies, share much the same biological and mental capabilities. The works of Boas, Lowie, Morgan, Tylor, Malinowski, and others demonstrated that little, if any, biological evolution has taken place in the human race in the past 1000 years. During the same period, however, considerable social revolution has occurred. Ogburn thus argued that a variable such as cultural change cannot be explained logically by a constant, biology. These facts mean that the mechanisms of social change can be found in the nature of culture itself, or in what Ogburn liked to term "the social heritage." Culture is unfortunately an umbrella term that includes all material objects, tools, artifacts, social institutions, customs, codes, and folkways.

Social evolution is a change in any of these elements. The moving force in evolution is the accumulation of cultural forms spurred by mechanical invention and scientific discovery.

Specifying the process in more detail, Ogburn divided culture into two parts: material and adaptive. Material culture—inventions, tools, mechanical processes, weapons, and other material items—changes first, while adaptive culture—the rest of social and institutional life—is “adapted” to changes in the material base. Ogburn also posited that material culture normally changes *faster*, while adaptive culture generally lags behind, seeking readjustment. It was not particularly clear why material changes ought to move faster, but Ogburn nevertheless believed that rapid changes in science and technology force lags to accumulate, increase social maladjustments, create disorganization, and bring about social problems [19, 23].

Later in the book *Social Change*, Ogburn specified the process of social evolution in greater detail by presenting the concepts of invention, accumulation, diffusion, and adjustment. With respect to invention, Ogburn wanted to show that mental ability, while necessary, is not sufficient to explain different levels of inventive activity. Mental ability cannot be the sole determinant of invention because numerous studies have suggested that individuals with the requisite mental ability have appeared in every society, even those with low inventive activity. Ogburn asserted, therefore, that economic demand also influences inventive activity by prompting larger investments in the inventive process and by influencing the actual application and use of new techniques. Furthermore, inventions are more abundant in large and diverse populations because many existing cultural elements can be recombined and reorganized in new and different ways [25].

Ogburn elaborated this view in a paper written with Dorothy Thomas: “Are Inventions Inevitable? A Note on Social Evolution” [30]. The authors asserted that biological and psychological attributes are secondary to cultural factors because many inventions and discoveries have been made independently by two or more persons. Ogburn and Thomas wrote that “if the various inventors had died in infancy, would not the inventions have been made and would not cultural progress have gone on without much delay?” (p. 85). This further supported the idea that “determinism in inventions [is] a matter of cultural preparation” (pp. 85–86). The central point, of course, was that social change is inherent in the nature of culture, not biology.

The second factor involved with social evolution—accumulation—refers to the condition when more new inventions are added to the culture base than are lost by obsolescence. If society or culture were equated with an accumulation of inventions, large accumulations would make additional inventions easier, implying that inventions should grow at exponential rates as existing techniques are recombined to produce new techniques. In the presence of large population increases, the rate of invention theoretically could be very great, although Ogburn recognized that, in reality, exponential growth usually lasts only briefly and takes on a cyclical character over time.

The third factor in the evolutionary process is diffusion, defined as the spread of inventions from one area to another. Diffusion indicates that specific groups are not required to make their own inventions because they can borrow them from other nations, population groups, or tribes. The availability of communication networks plays a significant part in the spread of invention. Ogburn suggested that nations living at communications crossroads can experience high levels of social evolution simply by borrowing a large variety of inventions from different sources. This theory also explains why geographically isolated areas have slower rates of social evolution. It does not explain, however, why some nations or groups are predisposed to borrow inventions more readily than others.

The final element of social evolution adjustment is perhaps the most familiar and controversial part of Ogburn's theory. Ogburn's initial assumption was that culture is a system; its parts are interrelated, and any change in one part results in changes in other elements that tend to lag behind. As Ogburn stated, "A cultural lag occurs when one or two parts of culture which are correlated changes before or in greater degree than the other part does, thereby causing less adjustment between the two parts than existed previously" [28, p. 86]. Cultural lag refers to the fact that adjustments do not take place automatically but are always made after some delay. This phenomenon, in turn, causes many problems for society. Ogburn believed these problems were becoming worse in the twentieth century because the cultural base was growing larger and the rate of invention was increasing rapidly and becoming overwhelming. He wrote in *Social Change* that cultural evolution was occurring "faster and faster until today mankind is almost bewildered in his effort to keep adjusted to these ever-increasing social changes" [19, pp. 199–200]. Such "bewilderment," however, resulted not just from strains in adaptive culture caused by technology but also from adjustment problems between *human* nature and technology. Indeed, Ogburn asserted that rapidly changing technology had social-psychological consequences such as alienation, powerlessness, mental illness, unhappiness, crime, and other ills [19, p. 287]. Thus Ogburn's theory provides, in addition, a rationale for understanding the causes of many forms of social disorganization and deviant behavior [13].

Later Extensions

Ogburn's work on technology and social evolution continued long after the publication of *Social Change*; it lasted well into the 1920s and 1930s, a period of rapid and profound social and economic change in the United States. Indeed, changes were so rapid and pervasive during this period that the U. S. government embarked on a systematic and comprehensive study of the major trends in American life. President Hoover reacted to these changes by creating the prestigious Research Committee on Social Trends in 1929. As research director of this committee, and as a sociologist, Ogburn occupied an influential position in this landmark recognition of the social sciences. Ogburn had also become a key member of the National Resources Committee, along with the noted political scientist Charles E. Merriam and the prominent economist Wesley Mitchell. The National Resources Committee reported its views in 1939 in the monumental study *Technological Trends and National Policy, Including the Social Implications of New Inventions* [18], sometimes considered the first modern technology assessment. This period of intensive government service provided Ogburn many opportunities to gather data, extend his theory, and test its application. The result was a more refined theoretical scheme than presented in the various editions of *Social Change*. These later extensions are often overlooked even though they represent an important element of Ogburn's thinking.

Perhaps the most important extension centered on the notion of social trends. Duncan [7, p. xvi] reminds us that Ogburn's studies of social evolution were not confined to the diagnosis of lags. A great deal of Ogburn's empirical study of social change was a collection of work more appropriately labeled a study of social trends. Ogburn had used trend lines in many of his earlier studies. The concept of social trend emerged by the late 1920s and early 1930s, when Ogburn was editor of *Social Changes*, an annual series of volumes for the *American Journal of Sociology*. Ogburn defined this concept in an unpublished manuscript cited by Duncan [7, pp. xvii–xviii]:

First, we need a paragraph on what social evolution is. We consider social evolution as not including, at least in the past several thousand years, any biological evolution. What is evolving then is society or the various social groupings such as government, industry, and the state. Expressed in other language, *what is evolving is "culture", that is to say, the environment which men have, but which the wild animals do not have. This culture, or our social heritage, is a composite of many different parts, such as cities, families, farms, philosophies, art, science, etc. Social evolution, or in other terms, cultural evolution may be broken down into the evolution of its parts.*

The evolution of the parts is designated as "social trends." Social trends are social movements over a considerable period of time in the same direction or one only slightly changing. This relative absence of violence of change in social trends gives a certain assurance of social stability. [Emphasis in original.]

In 1940, in Louis Wirth's edited volume commemorating the social sciences, *Eleven Twenty-Six: A Decade of Social Science Research* [40], Ogburn made it clear that the word *trend* also suggests measurement; trends cannot be discerned unless data are collected. Yet he recognized that measurement in lieu of other knowledge and theory is of little predictive value: "Hence it is not well to trust mechanically to measurement data alone, but it should be supplemented with other knowledge, even though it be not measurement. In this way prediction may be better improved" [p. 75]. The "other knowledge" Ogburn had in mind was careful describing, theorizing, and validating by the agreement of numerous observers [7, p. xvii]. The concept of social trends, then, had theoretical as well as methodological significance.

Ogburn's most important theoretical advance during this period was the assumption found at the heart of *Recent Social Trends*: There exist many social trends that are bound together and interrelated; together with invention, these trends push and pull against each other to move society through a kind of "dynamic equilibrium" in which new levels and states of adjustment are continuously sought and attained. This basically systemic view of society and its adjustment mechanisms has prompted some people to place Ogburn in the camp of equilibrium and system theorists [4, pp. 73–75]. This is a fair, if overlooked, classification. Consider Ogburn's review of the findings of *Recent Social Trends* (p. xiii):

It is the express purpose of this review of findings to unite such problems as those of economics, government, religion, education, in a comprehensive study of social movements and tendencies, to direct attention to the importance of balance among the factors of change. A nation advances not only by dynamic power, but by and through the maintenance of some degree of equilibrium among the moving forces.

The systems perspective of Ogburn's work in *Recent Social Trends* was further extended by his observation that movement seems to continue in the same general direction within the social system for long periods of time. A common pattern of social change is for scientific discoveries and inventions to instigate changes in the "economic organization and social habits which are most closely associated with them" [4, p. xiii]. Factories, cities, corporations, and labor organizations then grow up in response to technological developments. The next set of changes—perhaps better described as second-order effects—occur in organizations one step further removed, "namely in institutions such as the family, the government, the schools, and the churches" [4, pp. xii–xiv]. Such changes lead, in turn, to later modifications in social philosophies and codes of behavior. This sequence of change suggests a hierarchy of control and response within the different parts of the social system.

However, Ogburn recognized that changes do not always follow this sequence. He stressed that there must be a preexisting demand for an invention or it will not be used. Once an invention is introduced, it may be rejected or used only sparingly within society. Technical faults, the ready availability of substitutes, high production costs, state op-

position on social grounds (e.g., inventions increase the level of unemployment), vested interests that stand to lose relative advantages, and adverse popular opinion [20, p. 7] can retard the introduction of innovations. Thus Ogburn's theory came full circle: The social philosophies and attitudes of people can affect the nature and content of technological change, which eventually affects social institutions [11, p. 279]. Unfortunately, Ogburn never stated the conditions under which direction is reversed in this manner.

In the 1930s and 1940s, Ogburn continually revised these and earlier ideas. By 1937, when the National Resources Committee published *Technological Trends and National Policy* [18] under Ogburn's general direction and editorship, two new concepts had emerged in his theoretical scheme. The first concept was what Ogburn termed "primary influences," which referred to the fact that inventions have immediate effects on the persons or groups using them. Ogburn gave the following examples of primary influences [18, p. 9]:

If it be a producer's goods such as a farm tractor, it means at once the replacement of horses or mules, the purchasing of gasoline, and changes in various other farm practices. If it be a consumer's goods such as an air conditioning unit in a home, it affects the construction and use of the house, but of course the units must be fabricated and hence, for that purpose, factories must be created, marketing machinery set up, etc. All such results are called primary influences of the new technology.

The second concept involved "derivative effects," largely second-, third-, and higher-order changes produced by primary influences: "Thus as the tractor replaces animals on the farms there follows as a derivative influence less need for horse feed, which means that the land used for growing such feed is turned to other uses. This is a secondary effect" [18, p. 9]. Primary influences can have many derivative influences, each following the other like links in a chain, with successive derivative influences becoming smaller in magnitude. While the derivative influences spread themselves throughout society, primary effects of inventions are, in general, found in economic practices of production and in the habits of consumers using the finished product: "Very often then, the first general social institution affected by these changes has been the economic organization," Ogburn wrote [18, p. 9].

Ogburn expanded this line of reasoning in later writings by adding three additional concepts. The first concept, called "dispersion" in some of his writings and "diversification" in others [27], referred to the fact that a single invention often has multiple *primary* effects on many parts of society. Such effects fan out like spokes in a wheel to touch various areas of social life, institutions, and customs (see Figure 1). Drawing on data first presented in *Recent Social Trends* [20], Ogburn and Nimkoff [30, pp. 678–701] showed how radio's social effects were dispersed in at least 150 different areas.

A second concept, "succession," summarized the derivative social effects of inventions by showing that no given invention has ever influenced a single custom or social institution. Rather, impacts continue on and on, affecting additional social forms with declining intensity as effects are progressively played out (see Figure 2). Because impacts are dissipated as they move through the social structure, higher-order effects are difficult to detect and become almost invisible in many cases. This is an important insight, and it reflects the successive adaptations that social structures make in the face of technical advance. The idea suggests that change is slowly buffered and absorbed by numerous social forms rather than by a single unit which is forced to react all at once. The concept also implies that inventions may not directly "cause" certain changes, but they can have significant indirect effects nevertheless.

The final concept, "convergence," suggested that several different inventions can

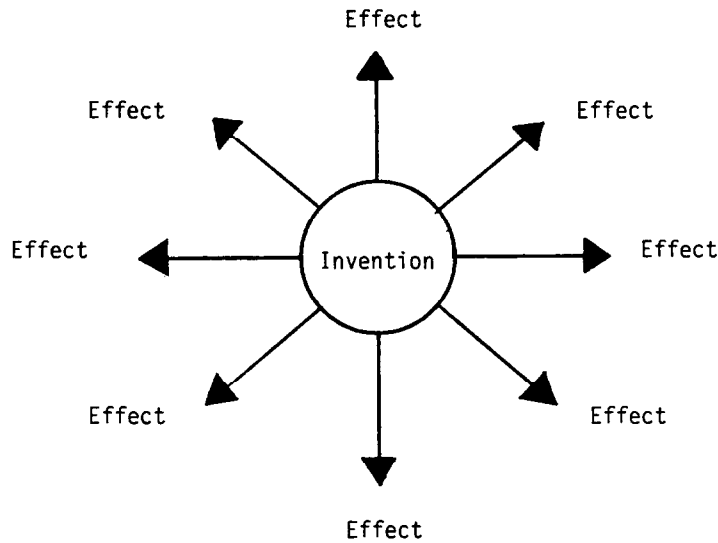


Fig. 1. Dispersion concept. A single invention often has multiple effects that fan out to touch many parts of society.

work together to produce a single primary effect (see Figure 3). The effects of a number of inventions sometimes accumulate in the same place, and Ogburn and Nimkoff [30, p. 703] demonstrated how declines in the birth rate resulted from increased use of factory machines, the rise of the automobile, contraceptives, compulsory education requirements, child labor laws, and other mechanical and social inventions. While the effect of any single invention may have been slight, convergence indicated that combined effects could work in synergistic fashion to create very appreciable social changes.

Regarding methodological considerations, Ogburn insisted that to predict the various effects of inventions, the researcher should draw projections based on the analysis of social trends—in essence, construct a social forecast. In his later work, Ogburn pursued this strategy in two ways. In one case, as we have seen, he tried to forecast the wide variety of impacts of a specific technology (dispersion); in the other, he tried to predict the likely future of a given area of social life in view of projected technological changes. In the first case, Ogburn used “prediction” based on measurement and quantitative data wherever possible; for example, over 50 variables were used in trend analyses and extrapolations to forecast the impact of aviation [26]. Yet, because he was well aware of the problems of quantifying effects of future technologies, Ogburn also stressed the

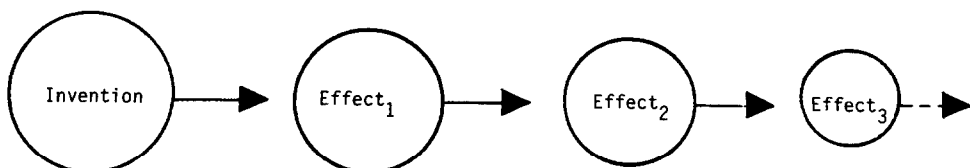


Fig. 2. Succession concept. The impacts of an invention decline in intensity as they move through the social structure.

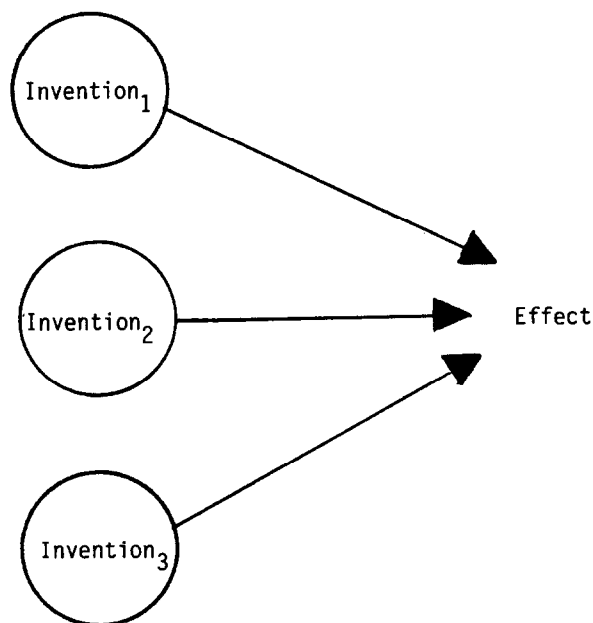


Fig. 3. Convergence concept. Several different inventions can work together to produce a single primary effect.

usefulness of drawing up lists of possible effects (recall the 150 effects listed for the radio), even if these were not measured in degree or time.

In the second case, where Ogburn looked at the future of a given area of life or a social institution such as the family, changes in the institution were taken as dependent variables, with technology the independent variable producing the changes. The method followed was, first, to locate and describe trends in the past evolution of the family. In part adumbrating the Delphi technique developed many years later by mathematicians at the RAND Corporation, Ogburn and Nimkoff's [29] study of technology and the American family used expert judgment to locate significant trends occurring in the family over the previous century. The resulting list of changes was compressed into eight interrelated trends such as earlier marriages, increases in the proportion of working wives, increasing pressure toward equalizing the authority of husband and wife, and more emphasis on romantic love. These previous trends were then related to changes in the independent variable (technology), and the relationships obtained were used to project the future course of events in family life given further modifications in the independent variable. This type of social forecasting was actually far ahead of its time, and it resembles many recent forecasting techniques (for a related discussion, see [23]).

It is remarkable how accurate many of Ogburn's forecasts proved to be. He was essentially correct in his anticipation of the widespread adoption of television, the proliferation of artificial fibers, and the development of oral contraceptives, among others. Yet the possibility of increased social control of technological advance tended to be overshadowed by his discussions of adaptations in life styles, culture, and behavior produced by technological change. Again, this situation arose out of Ogburn's use of quantification and trend analysis which downplayed the significance of qualitative shifts and changes in societal value orientations. Still, Ogburn's predictions and forecasts were far more

informed and sophisticated than most latter-day technological “fixers” or determinists. He was absolutely right to insist that social planning and policy making, if they were to be effective, should include forecasts about the future. In short, the question “What is likely to happen?” had to be asked and at least tentatively answered.

Response and Critique

Ogburn's theories of technology and social change—both earlier versions and later extensions—were not without their critics. Most criticism of his earlier work centered on the notions of cultural lag and adjustment. For example, Ogburn did relatively little to explain the nature of the delays involved in social adjustment to technological change, nor did he tell much about the factors that influenced the length of these delays. In addition, he did not expand on the differences between adjustments *within* levels of the social system as opposed to those *between* levels. In fact, because Ogburn never considered cultural lag and adjustment the major elements of this theory, he did not say much about these concepts beyond the original formulation in *Social Change* [11].

Nevertheless, critics focused on the lag thesis and accused Ogburn of proffering a “technological interpretation of history” similar to the Marxian position, if not a modern reformulation of it. The criticism contended that the theories of material and adaptive culture were little more than restatements of Marx's substructure and superstructure concepts. Such a technological interpretation of social change makes the initial error of failing to see that objects like technology are only one part of culture. The emphasis on material culture leaves the impression that machines are self-inventing, self-perpetuating, and self-expanding [16, p. 549], a position which places Ogburn alongside such technological determinists as Jacques Ellul [10]. Technology *may* be more susceptible to change than other culture segments and, therefore, may possibly be causally significant to social change, if the following conditions are present: 1) technological progress is assumed and remains constant, that is, technological progress is the goal, and 2) the relationship between different societal values must be subject to variation, since technology is capable of expansion only to the degree that the consequences of modifying other values can be absorbed and accepted by society. Expanding industrial technology continues to be a causal factor precisely because society has been willing to make the necessary adjustments [16, p. 461]. While inventions may have social results, they also have social causes, and their rejection or acceptance depends on the social framework. Cultural lag may result from unanticipated consequences of technological advance, but the lag is not a predestined result of the slowness of adaptive culture in catching up; the lag is capable of purposive solution [16, p. 462].

Perhaps the more serious criticism was aimed not at Ogburn's technological interpretation of social change, but at the specific analytical and empirical aspects of the theory. Critics frequently attacked Ogburn on the grounds that the terms *material culture* and *adaptive culture* suffered from a lack of operational definition. Material culture, for instance, could be taken to mean almost anything: science, technology, technique, invention, natural resources, transportation, or manufacturing. In addition, material culture was not sufficiently distinct from adaptive culture, for technique itself is but a form of science and knowledge. Any technical device is an incorporation of knowledge concerning that device and, as such, is often conditioned by the general state of science in that and related fields. Technical knowledge, therefore, cannot be readily distinguished from other forms of knowledge [36, pp. 304–305].

Perhaps more poignant was Sorokin's [37] belief that Ogburn and his followers could never prove convincingly that the pace of change in material culture and invention is far

greater than in other sociocultural phenomena because the proper unit of velocity by which to measure this pace could not be determined. This problem arises from the fact that most social change is qualitative rather than quantitative. Social life is heterogeneous and irreducible to a single, common, quantitative denominator. It is impossible to "measure" the comparative speeds of different parts of material and adaptive culture to tell what is lagging behind what. Expressed another way, what lags behind what depends on one's initial point of reference [39]. Similarly, there is no way to predict beforehand which changes will take the longest periods of adjustment. For example, would the introduction of tractors require longer or more sweeping adjustments than the introduction of the airplane? Moreover, are changes in material culture necessarily disintegrative [36]?

Some of the later extensions of Ogburn's theory have been subjected to similar criticisms. The concepts of primary and derivative effects, for example, were never operationally defined. How does one specify an "effect"? What are the indicators of effects? How does one determine whether some effects are greater than others? Ogburn gave many examples of such effects but never expressed them in systematic theoretical language. Ogburn's theory, moreover, never showed how to determine whether the effect in question is actually the result of a given invention rather than of a variable not included in the model. The problem of how one apportions the individual effects of several inventions, as in the case of convergence, was never specifically addressed, nor was the question of the differential impact of inventions. Finally, later extensions discounted the fact that the environment into which a new technology is dispersed always presents a preexisting structure to channel impacts. Existing states of equilibrium have selective influences on how and where the effects of inventions will be dispersed.

The core of these problems reverts to Ogburn's earlier work on cultural lag. The key issue, as with the lag concept, is to define the nature and meaning of correlation between social elements by way of at least an implicit theory of society which specifies the ways in which various parts or subsystems are interrelated. Ogburn's work on social trends suggested such interrelations, but did not identify their nature in detail. Perhaps it is more accurate to say that Ogburn had certain theoretical orientations rather than a theory of society, *per se*. Had Ogburn articulated more clearly the systems perspective inherent in his approach and demarcated the relations between systems and subsystems, many of his later refinements concerning how invention causes social change would have been improved considerably. Ogburn's approach is, in the end, syndromic, leading to comparisons of slices *in* time. Yet the trend concept is clearly diachronic, projecting events *through* time [9]. Consequently, the question which continually plagued Ogburn's analysis, and which he never faced squarely, was how to combine these contrasting approaches to explain and predict social changes resulting from technical advance. The concepts of primary and derivative effects, dispersion, succession, and convergence are highly useful, but they fail to obviate these fundamental difficulties.

On the methodological front, perhaps the sharpest criticisms concern Ogburn's use of trend analysis and extrapolation. One of the central problems with extrapolation of trends based on quantitative data is the failure to account for changes caused by shifting values. For example, the collection of data on commercial air travel in the early twentieth century would have indicated a trend toward increasing speed. (Ogburn himself, incidentally, was rather pessimistic about the use of jet aircraft and overoptimistic regarding the use of the helicopter.) Yet few forecasters could have accounted for the surprise shift in values forced into the political arena years later by environmentalists who opposed supersonic transport aircraft, thereby halting this trend. Had Ogburn thought about such opposition, it presumably would have been residualized as a cultural lag awaiting the

necessary adjustments. Similarly, a 1940s trend analysis of energy projections probably would not have predicted the nuclear power controversy in the 1970s. Much of the debate can be traced to the shift in values and ideologies of antitechnologists and environmental lobbyists who had gained increasing influence in societal decision making [6, 8, 15].

Another problem encountered by trend analysis, particularly when predicting the effects of new inventions, is the frequent lack of adequate quantitative data and historical evidence to validate judgments and interpretations. Even when good data are available, social change is rarely a smooth linear progression devoid of countertrends and the probability that the forces producing change will not wane in the future. There is likewise always the danger that embryonic, yet possibly significant, trends will not be spotted because they appear inconsequential at the time of the analysis. For example, Ogburn and Nimkoff's *Technology and the Changing Family* [29] predicted the increasing use of mobile homes. These trailers were expected to serve as accommodations for special family arrangements, however—such as those who are very mobile or childless—rather than as principal residences for conventional nuclear families, as is now frequently the case. The unforeseen factors affecting the validity of this forecast were record-high interest rates and high prices of conventional single-family dwellings. Even the most carefully constructed trend analysis cannot be projected too far into the future without being subject to such unanticipated events.

Reconsideration and Extension

Perhaps the most fruitful way to reconsider Ogburn's ideas is from a systems approach. One of Ogburn's key insights was the fashion in which technology's effects move through society: invention and technology affect, first, the economic organizations and social structures most closely associated with economic activity, with effects moving next through family, government, education, and religion, followed by shifts in ideologies, philosophies, and social codes of behavior—i.e., changes in the symbolic universe. This certainly can be compared to other systems perspectives such as that offered by Talcott Parsons in his cybernetic hierarchy of control, where changes are seen moving up through various levels or subsystems of society to major value orientations [32]. While Ogburn did not necessarily see the symbolic universe in a "control" position as did Parsons, he nonetheless tried to explain why technology's effects on values and philosophies are often slight and hard to detect: By the time technological impacts reach the symbolic universe, they are dissipated by the successively weaker impacts of derivative effects. Another way of expressing this concept is that technology never alters ideas and values as a primary effect, but rather works first through other institutions and social forms that absorb and dilute primary impacts. Ogburn might have objected to this reformulation, but there is little question that his ideas can be considered from such a perspective. A systems interpretation might be seen as a fruitful alternative to the technological primacy label usually placed on his work, and new extensions might proceed via this approach.

Other significant contributions that deserve extension concern Ogburn's views of technology assessment and predicting the social effects of invention. His concepts of primary and derivative effects, including the processes of dispersion, succession, and convergence, are insightful and rich in hypotheses about how technology causes social change. The National Academy of Sciences Panel on Technology Assessment [17] has shown that the nature of most technological impacts is distributed throughout society in something like the first, second, third progression suggested by Ogburn. Some of the failings of current-day technology assessments result from the lack of adequate social theory and basic assumptions about the nature of society and how it functions, something

Ogburn was careful to include in nearly all his studies of technology. We should be willing to return to Ogburn's basic approach even if certain elements of his theory are rejected.

As for the connection of Ogburn's work to policy making, the concept of social trends is still highly relevant. No policy can be effective without a set of assumptions about social dynamics and a forecast about the future. Ogburn was entirely right to suggest the continuing importance of collecting information on such trends and to insist that this task be institutionalized so that social science would have a firm place in the government's policy-making apparatus. This was a reasonable alternative to the reformist, technocratic, and muckraking social analysis of his own time [1] as well as a rejoinder to those who would eschew the importance of social science in contemporary forecasting. Predicting the social effects of invention is too complex a task to be left to one or a few fields; it must be a pluralistic effort undertaken by the full range of disciplines. Thus, the concept of social trends remains a useful tool that should be extended and refined by specialists from many disciplines.

Extensions of Ogburn's work in other areas might also be continued. The impact of technology on particular institutional spheres should be moved forward, as should the research on the diffusion of innovations [2, 35]. Similarly, Price's work on the accumulation of knowledge [34] and Merton's line of inquiry on multiple invention [14] offer additional areas for further research. Some of the most interesting work drawing on Ogburn's general approach has been conducted under the heading of "social indicators" [5] and "technological forecasting" [15]. Both kinds of research follow the tradition set forth by Ogburn and his associates in *Recent Social Trends* and *Technological Trends and National Policy*, and work in these important areas can profit from reconsideration and extension of Ogburn's theories and methods of studying technology and social evolution.

Conclusion

William Fielding Ogburn died on April 27, 1959, in Tallahassee, Florida. The *New York Times*, for which he had so often written articles and editorials, hailed him as "an educator in many fields who had compiled the first national inventory for government" (*Recent Social Trends*). He was a leading social scientist, critic, and social philosopher who held an influential position in the academic and government communities for nearly 40 years. He left a broad legacy of vast contributions to sociology, economics, political science, and other fields. Yet throughout this diversity runs the common theme of technology and social change. Ogburn saw the great significance of technology and did more than anyone else to explore its social effects.

Today, the pace of technological advance is every bit as rapid as in Ogburn's time. The increasing emphasis on science and technology, the ascendancy of experts and theoretical knowledge, and other contemporary developments suggest that social adjustments to technological change will be many. It is little wonder that technology assessment proceeds with great vigor in many quarters—government, business, and the academic community, among others. Yet, with all this activity exploring the social effects of technology, too little attention is afforded theoretical work concerning the nature of the social order and the precise ways in which technology produces social change. Improved and more accurate assessments call for better theoretical models, concepts, and operational definitions. Ogburn initiated this endeavor for us, and we should pick it up with new energy and move it forward once again.

The author thanks Harold Pfautz for several readings of earlier versions of the manuscript. Otis D. Duncan and A.J. Jaffee also read earlier versions and provided many helpful insights. A. Hunter Dupree and Mark Shields made several substantive and stylistic suggestions, and their comments are gratefully acknowledged.

References

1. Aiken, William E., *Technocracy and the American Dream: The Technocratic Movement, 1900–1941*, University of California Press, Berkeley, 1977.
2. Allen, Francis R. et al., *Technology and Social Change*, Appleton-Century-Crofts, New York, 1957.
3. Allen, Frederick Lewis, *Only Yesterday: An Informal History of the 1920s*, Harper and Brothers, New York, 1931.
4. Appelbaum, Richard P., *Theories of Social Change*, Markham, Chicago, 1970.
5. Bauer, Raymond A., ed., *Social Indicators*, MIT Press, Cambridge, Mass., 1966.
6. Del Sesto, Steven L., Conflicting Ideologies of Nuclear Power: Congressional Testimony on Nuclear Reactor Safety, *Public Policy* 28(Winter), 39–70 (1980).
7. Duncan, Otis Dudley, ed., *On Culture and Social Change*, University of Chicago Press, Chicago, 1964.
8. Duncan, Otis Dudley, Sociologists Should Reconsider Nuclear Energy, *Social Forces* 57(September), 1–22 (1978).
9. Dupree, A. Hunter, Personal Communication to the Author, March 3, 1982.
10. Ellul, Jacques, *The Technological Society*, Knopf, New York, 1964.
11. Jaffe, A. J., "Ogburn, William Fielding," in *International Encyclopedia of Social Sciences* (Vol. 11), Macmillan, New York, 1968.
12. Jaffe, A. J., Personal Communication to the Author, July 7, 1981.
13. Martindale, Don, *The Nature and Types of Sociological Theory*, Houghton-Mifflin, Boston, 1960.
14. Merton, Robert K., *The Sociology of Science: Theoretical and Empirical Investigations*, University of Chicago Press, Chicago, 1973.
15. Miles, Ian, *The Poverty of Prediction*, D.C. Heath, Lexington, Mass., 1975.
16. Moore, Wilbert E., The Sociology of Economic Organization, in *Twentieth Century Sociology*. Georges Gurwitsch and William E. Moore, eds., The Philosophical Library, New York, 1945.
17. National Academy of Sciences Panel on Technology Assessment, *Technology: Processes of Assessment and Choice*, U.S. Government Printing Office, Washington, D.C., 1968.
18. National Resources Committee, *Technological Trends and National Policy, Including the Social Implications of New Inventions*, U.S. Government Printing Office, Washington, D.C., 1937.
19. Ogburn, William, *Social Change, with Respect to Culture and Original Nature*, Viking Press, New York, 1922.
20. Ogburn, William, Introduction, in *Recent Social Trends in the United States*. President's Research Committee on Social Trends, McGraw-Hill, New York, 1933.
21. Ogburn, William, National Policy and Technology, in *Technological Trends and National Policy, Including the Social Implications of New Inventions*. Subcommittee on Technology, U.S. Government Printing Office, Washington, D.C., 1937.
22. Ogburn, William, Social Trends, in *Eleven Twenty-Six: A Decade of Social Science Research*. L. Wirth, ed., University of Chicago Press, Chicago, 1940.
23. Ogburn, William, Technology and Planning, in *Planning for America*. G. Galloway, ed., Holt, New York, 1941.
24. Ogburn, William, Technology as Environment, *Sociology and Social Research* (September–October), 3–9, 1945.
25. Ogburn, William, Inventions of Local Transportation and the Patterns of Cities, *Social Forces* 24(May) 373–79, 1946.
26. Ogburn, William, *The Social Effects of Aviation*, Houghton-Mifflin, Boston, 1946.
27. Ogburn, William, Social Effects of Technology in Industrialized Societies, *International Social Science Bulletin* 4(Summer) 269–79, 1952.
28. Ogburn, William, Cultural Lag as Theory, in *Culture and Social Change*. O. D. Duncan, ed., University of Chicago Press, Chicago, 1964.
29. Ogburn, William, and Nimkoff, Meyer, *Technology and the Changing Family*, Houghton-Mifflin, Boston, 1955.
30. Ogburn, William, and Nimkoff, Meyer, *Sociology*, Houghton-Mifflin, Boston, 1964.
31. Ogburn, William, and Thomas, Dorothy, Are Inventions Inevitable? A Note on Social Evolution, *Political Science Quarterly* 37(March), 83–98 (1922).

32. Parsons, Talcott, *Social Systems and the Evolution of Action Theory*, Free Press, New York, 1977.
33. President's Research Committee on Social Trends, *Recent Social Trends in the United States*, McGraw-Hill, New York, 1933.
34. Price, Derek, *Science since Babylon*, Yale University Press, New Haven, Conn., 1961.
35. Rogers, Everett M., *Communication of Innovations*, Free Press, New York, 1971.
36. Sorokin, Pitrim A., *Contemporary Sociological Theories*, Harper and Row, New York, 1928.
37. Sorokin, Pitrim A., Recent Social Trends: A Criticism, *Journal of Political Economy* 41(April), 194–210, 1933.
38. Sorokin, Pitrim A., *Social and Cultural Dynamics*, American Book Co., New York, 1941.
39. Wallis, Walter A., The Concept of Lag, *Sociology and Social Research* 19(May–June), 303–305, 1935.
40. Wirth, Louis, ed., *Eleven Twenty-Six: A Decade of Social Science Research*, University of Chicago Press, Chicago, 1940.

Received 6 September 1982; revised 13 January 1983